

QUANTUM OPTIMAL TRANSPORT ON BOST–CONNES SYSTEMS: THE TWIST IS HARMFUL, THE GENERATOR HAS A SIGN ERROR, AND THE TRANSITION IS INVISIBLE TO LOCAL INEQUALITIES

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ABSTRACT. We examine three proposals for a quantum optimal transport theory on the type III Bost–Connes systems and find that each fails, in each case for an identifiable and correctable reason. Two of the corrections are clean; the third is a structural obstruction.

The modular twist is not merely unnecessary but harmful. With $D_{\text{vert}}\epsilon_a = (\log Na)\epsilon_a$ and $\vartheta = \sigma_{-i\beta}$, one computes

$$[D_{\text{vert}}, \mu_p]_{\vartheta} \epsilon_b = (\log Np + (1 - Np^\beta) \log Nb) \epsilon_{pb},$$

which is *unbounded*. The untwisted commutator is bounded and elementary: $D_{\text{vert}}\mu_p = \mu_p(D_{\text{vert}} + \log Np)$ gives $[D_{\text{vert}}, \mu_p] = (\log Np)\mu_p$. The reason is that a twist repairs a *multiplicative* mismatch, whereas D_{vert} is *additive* in $\log N$; introducing ϑ manufactures the mismatch it was meant to cure. The untwisted $D_{\text{full}} = D_{\text{base}} \otimes \sigma_1 + D_{\text{vert}} \otimes \sigma_2$ does have $\ker L = \mathbb{C}1$ — and, as it turns out, exactly when S is norm-separated, the same condition that governs the equal-norm analysis of Paper XIV of the series (its Theorem 1.3). Whether it is a spectral triple reduces to whether the horizontal operator has compact resolvent, which is not established.

The proposed Lindbladian does not preserve the KMS state. Writing a_p, b_p for the rates of $\mu_p^*(\cdot)\mu_p$ and $\mu_p(\cdot)\mu_p^*$, the KMS condition gives $\varphi(\mathcal{L}f) = \sum_p \{\nu(fe_p)(a_p Np^\beta - b_p) + \nu(f)(b_p Np^{-\beta} - a_p)\}$, which vanishes for all f if and only if $b_p = a_p Np^{+\beta}$. The proposal takes $b_p = a_p Np^{-\beta}$: *the sign of the exponent is wrong*, and with it $\varphi(\mathcal{L}e_p) = (1 - t^2)(1 - t) \neq 0$ for $t = Np^{-\beta}$. The corrected ratio $b_p/a_p = e^{\beta \log Np}$ is the detailed-balance factor, as it must be.

Critical slowing down does not occur. The proposed asymptotic $\alpha(\beta) \asymp \Pi_\infty(\beta) = \prod_\ell \frac{1 - \ell^{-\beta}}{1 + \ell^{-\beta}}$ is of the wrong shape: spectral gaps, modified log-Sobolev constants and Bakry–Émery curvature bounds *tensorise to an infimum* of single-site quantities, each a function of its own $\ell^{-\beta}$ alone, not to a product over ℓ . The single-site defects $1 - \ell^{-\beta}$ are bounded below by $1 - 3^{-\beta} \geq 0.66$, and the single-site spectral gap by $(1 - t)^2/4t$ (the Poincaré constant of the geometric law), so with unit rates the gap stays bounded below near β_c while $\Pi_\infty \rightarrow 0$. The arithmetic transition is a *tail* phenomenon — the Kakutani singularity of an infinite product — and such tensorising local quantities cannot see it. This is the same mechanism that made the metric transition of Paper IV of the series (its Theorem 6.2) continuous rather than first-order.

1. THE TWIST

We use the notation of [5]: $\mathcal{A}_{\mathbb{Q}, S} = C(Y_S) \rtimes J_S$, spanned by the monomials $\mu_a f \mu_b^*$, with $\sigma_t(\mu_a) = Na^{it}\mu_a$; X and ν_{sym} are the compact model and its symmetric measure. Papers III, IV and XIV of the series, referred to by numeral, are unpublished companions; every fact used from them is restated or re-derived in place. Represent $\mathcal{A}_{\mathbb{Q}, S}$ on $\mathcal{H}_0 = L^2(X, \nu_{\text{sym}}) \otimes \ell^2(J_S)$ in the Toeplitz representation, $\mu_a \epsilon_b = \epsilon_{ab}$, and set $D_{\text{vert}}\epsilon_a = (\log Na)\epsilon_a$. Let $\vartheta = \sigma_{-i\beta}$, so $\vartheta(\mu_a) = Na^\beta \mu_a$ and $\vartheta(f) = f$ for $f \in C(Y_S)$.

Proposition 1.1 (The twisted commutator is unbounded). *For every prime p and every $\beta > 0$,*

$$(1.1) \quad [D_{\text{vert}}, \mu_p]_{\vartheta} \epsilon_b = (D_{\text{vert}}\mu_p - \vartheta(\mu_p)D_{\text{vert}})\epsilon_b = (\log Np + (1 - Np^\beta) \log Nb) \epsilon_{pb},$$

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in [papers/XVI-transport/](#).

whose coefficient tends to $-\infty$ as $N\mathbf{b} \rightarrow \infty$. Hence $[D_{\text{vert}}, \mu_{\mathbf{p}}]_{\vartheta} \notin \mathcal{B}(\mathcal{H}_0)$.

Proof. $D_{\text{vert}}\mu_{\mathbf{p}}\epsilon_{\mathbf{b}} = \log N(\mathbf{p}\mathbf{b})\epsilon_{\mathbf{p}\mathbf{b}} = (\log N\mathbf{p} + \log N\mathbf{b})\epsilon_{\mathbf{p}\mathbf{b}}$ and $\vartheta(\mu_{\mathbf{p}})D_{\text{vert}}\epsilon_{\mathbf{b}} = N\mathbf{p}^{\beta}(\log N\mathbf{b})\epsilon_{\mathbf{p}\mathbf{b}}$; subtract. Since $N\mathbf{p}^{\beta} \neq 1$ the coefficient is unbounded in $\mathbf{b}$. $\square$

Proposition 1.2 (The untwisted commutator is bounded). $D_{\text{vert}}\mu_{\mathbf{a}} = \mu_{\mathbf{a}}(D_{\text{vert}} + \log N\mathbf{a})$, hence

$$[D_{\text{vert}}, \mu_{\mathbf{a}}] = (\log N\mathbf{a})\mu_{\mathbf{a}}, \quad \|[D_{\text{vert}}, \mu_{\mathbf{a}}]\| = \log N\mathbf{a} < \infty,$$

and $[D_{\text{vert}}, f] = 0$ for $f \in C(Y_S)$.

Proof. Both sides send $\epsilon_{\mathbf{b}}$ to $(\log N\mathbf{a} + \log N\mathbf{b})\epsilon_{\mathbf{a}\mathbf{b}}$; the vanishing on $C(Y_S)$ is the observation recorded in Paper IV of the series (its Proposition 2.1), D_{vert} being diagonal in a basis with respect to which $C(Y_S)$ acts diagonally. $\square$

Remark 1.3 (Why the twist backfires). A Connes–Moscovici twist [3] repairs a *multiplicative* discrepancy: it is used when conjugation by the algebra rescales D , so that $Dx - xD$ fails to be bounded by a factor, and ϑ absorbs the factor. Here D_{vert} is *additive* in $\log N$, so $Dx - xD$ is already bounded, and inserting the multiplicative $\vartheta(\mu_{\mathbf{a}}) = N\mathbf{a}^{\beta}\mu_{\mathbf{a}}$ introduces a factor where there was none. Equation (1.1) is precisely the manufactured mismatch. The twisted triple, with twist $\sigma_{i\beta}$, was the candidate proposed in the closing assessment of Paper IV; this section is the reason we do not pursue it.

Remark 1.4 (Numerical illustration). For $N\mathbf{p} = 3$, $\beta = 1$ the coefficient in (1.1) takes the values $-3.5, -12.7, -26.5, -54.2$ at $N\mathbf{b} = 10, 10^3, 10^6, 10^{12}$; for $\beta = 2$, $-17.3, -54.2, -109.4, -219.9$. The untwisted commutator has constant norm $\log 3 = 1.0986$.

2. THE UNTWISTED OPERATOR, AND THE REAPPEARANCE OF NORM-SEPARATION

Let $D_{\text{full}} = D_{\text{base}} \otimes \sigma_1 + D_{\text{vert}} \otimes \sigma_2$ on $\mathcal{H} = \mathcal{H}_0 \otimes \mathbb{C}^2$, with D_{base} the horizontal operator of Paper IV (its Definition 2.4), amplified to act on the $L^2(X, \nu_{\text{sym}})$ factor of every summand $L^2(X, \nu_{\text{sym}}) \otimes \epsilon_{\mathbf{b}}$, and $L(x) = \|[D_{\text{full}}, x]\|$. In this representation $\mu_{\mathbf{a}}$ acts on $\ell^2(J_S)$ alone and $f \in C(Y_S)$ acts on the summand indexed by $\mathbf{b}$ as multiplication by the translate $f \circ m_{\mathbf{b}}$, so $[D_{\text{base}}, \mu_{\mathbf{a}}] = 0$ while $[D_{\text{base}}, f]$ is the $\mathbf{b}$ -indexed family $[D_{\text{base}}, f \circ m_{\mathbf{b}}]$.

Proposition 2.1 (Kernel). *The operators σ_1, σ_2 anticommute and x acts diagonally, so $[D_{\text{full}}, x] = [D_{\text{base}}, x] \otimes \sigma_1 + [D_{\text{vert}}, x] \otimes \sigma_2$, which vanishes iff both terms do. By Proposition 1.2, $\ker[D_{\text{vert}}, \cdot] = \mathcal{A}_{\mathbb{Q}, S}^{\sigma}$. Hence*

$$\ker L = \{x \in \mathcal{A}^{\sigma} : [D_{\text{base}}, x] = 0\},$$

which equals $\mathbb{C}1$ if and only if $\mathcal{A}^{\sigma} = C(Y_S)$, i.e. if and only if S is norm-separated in the sense of Theorem 1.3 of Paper XIV.

Proof. $[D_{\text{vert}}, \mu_{\mathbf{a}}f\mu_{\mathbf{b}}^*] = (\log N\mathbf{a} - \log N\mathbf{b})\mu_{\mathbf{a}}f\mu_{\mathbf{b}}^*$, zero iff $N\mathbf{a} = N\mathbf{b}$, so the kernel of the vertical commutator is exactly the fixed-point algebra of σ . For $f \in C(Y_S)$, $[D_{\text{base}}, f] = 0$ forces $[D_{\text{base}}, f \circ m_1] = 0$ on the summand $\mathbf{b} = (1)$, hence $f \in \mathbb{C}1$ by the Lip-kernel lemma of Paper IV (its Lemma 3.1); if $\mathcal{A}^{\sigma} \supsetneq C(Y_S)$ the element $\mu_{\mathbf{a}}\mu_{\mathbf{b}}^*$ with $N\mathbf{a} = N\mathbf{b}$, $\mathbf{a} \neq \mathbf{b}$, lies in $\ker[D_{\text{vert}}, \cdot]$ and, acting on $\ell^2(J_S)$ alone, also in $\ker[D_{\text{base}}, \cdot]$. $\square$

Remark 2.2 (Spectral triple: not established). Since D_{base} and $1 \otimes D_{\text{vert}}$ commute and are placed on anticommuting Clifford directions, $D_{\text{full}}^2 = D_{\text{base}}^2 \otimes 1 + 1 \otimes D_{\text{vert}}^2$, and on the summand $L^2(X, \nu_{\text{sym}}) \otimes \epsilon_{\mathbf{a}}$ this is $D_{\text{base}}^2 + (\log N\mathbf{a})^2$. Over $\mathbb{Q}$ the norms $N\mathbf{a}$ are distinct and tend to infinity, so D_{full} has compact resolvent if and only if D_{base} does on $L^2(X, \nu_{\text{sym}})$ — a property of the weight sequence of Paper IV, which that paper does not establish and we do not claim; its own assessment leaves the spectral triple on the crossed product open, and so do we. In particular no index theory is asserted. The pair $(\mathcal{A}_{\mathbb{Q}, S}, L)$ can still be a compact quantum metric space in Rieffel’s sense [4] —

which requires only that the Lipschitz ball be totally bounded modulo scalars — and it is that notion, not the spectral triple, that is at issue here; the two are often conflated.

Remark 2.3 (Open Problem 1 of Paper IV). Proposition 2.1 settles the kernel question but not the compact quantum metric space question, which additionally requires total boundedness of $\{x : L(x) \leq 1, \varphi(x) = 0\}$. The vertical directions have Lipschitz cost bounded below by $\log N\mathfrak{p}_0$ for the smallest prime, which is encouraging; we have not verified total boundedness and do not claim it.

3. THE GENERATOR: A SIGN ERROR IN THE DETAILED BALANCE

Proposition 3.1 (Invariance forces the detailed-balance ratio). *Let*

$$\mathcal{L}(x) = \sum_{\mathfrak{p} \in S} a_{\mathfrak{p}} (\mu_{\mathfrak{p}}^* x \mu_{\mathfrak{p}} - x) + \sum_{\mathfrak{p} \in S} b_{\mathfrak{p}} \left(\mu_{\mathfrak{p}} x \mu_{\mathfrak{p}}^* - \frac{1}{2} (e_{\mathfrak{p}} x + x e_{\mathfrak{p}}) \right), \quad e_{\mathfrak{p}} = \mu_{\mathfrak{p}} \mu_{\mathfrak{p}}^*,$$

the GKLS form with jump operators $\mu_{\mathfrak{p}}$ and $\mu_{\mathfrak{p}}^$. Then $\mathcal{L}(1) = 0$, and for $f \in C(Y_S)$ — where $\frac{1}{2}(e_{\mathfrak{p}} f + f e_{\mathfrak{p}}) = e_{\mathfrak{p}} f$ — the KMS condition $\varphi(xy) = \varphi(y \sigma_{i\beta}(x))$ gives $\varphi(\mu_{\mathfrak{p}}^* x \mu_{\mathfrak{p}}) = N\mathfrak{p}^{\beta} \varphi(x e_{\mathfrak{p}})$ and $\varphi(\mu_{\mathfrak{p}} x \mu_{\mathfrak{p}}^*) = N\mathfrak{p}^{-\beta} \varphi(x)$ — the scaling relations recorded as Proposition 5.1 of Paper III of the series — whence*

$$(3.1) \quad \varphi(\mathcal{L}(f)) = \sum_{\mathfrak{p}} \left\{ \nu(f e_{\mathfrak{p}}) (a_{\mathfrak{p}} N\mathfrak{p}^{\beta} - b_{\mathfrak{p}}) + \nu(f) (b_{\mathfrak{p}} N\mathfrak{p}^{-\beta} - a_{\mathfrak{p}}) \right\}.$$

This vanishes for every f if and only if

$$(3.2) \quad b_{\mathfrak{p}} = a_{\mathfrak{p}} N\mathfrak{p}^{+\beta} \quad \text{for every } \mathfrak{p} \in S.$$

Proof. $\mathcal{L}(1) = \sum a_{\mathfrak{p}} (\mu_{\mathfrak{p}}^* \mu_{\mathfrak{p}} - 1) + \sum b_{\mathfrak{p}} (e_{\mathfrak{p}} - e_{\mathfrak{p}}) = 0$. The two scaling relations are the KMS condition with $y = \mu_{\mathfrak{p}}$, respectively $y = \mu_{\mathfrak{p}}^*$, and $\sigma_{i\beta}(\mu_{\mathfrak{p}}) = N\mathfrak{p}^{-\beta} \mu_{\mathfrak{p}}$. For (3.1) apply them termwise. If (3.2) holds, both brackets vanish. Conversely write $A_{\mathfrak{p}} = a_{\mathfrak{p}} N\mathfrak{p}^{\beta} - b_{\mathfrak{p}}$ and $B = \sum_{\mathfrak{p}} (b_{\mathfrak{p}} N\mathfrak{p}^{-\beta} - a_{\mathfrak{p}})$; under ν the valuations are independent with $\nu(e_{\mathfrak{p}}) = t_{\mathfrak{p}} = N\mathfrak{p}^{-\beta}$ and $\nu(e_{\mathfrak{p}} e_{\mathfrak{q}}) = t_{\mathfrak{p}} t_{\mathfrak{q}}$ for $\mathfrak{p} \neq \mathfrak{q}$, so $f = 1$ gives $\sum_{\mathfrak{p}} A_{\mathfrak{p}} t_{\mathfrak{p}} + B = 0$ while $f = e_{\mathfrak{q}}$ gives $t_{\mathfrak{q}} (A_{\mathfrak{q}} + \sum_{\mathfrak{p} \neq \mathfrak{q}} A_{\mathfrak{p}} t_{\mathfrak{p}} + B) = 0$; subtracting, $A_{\mathfrak{q}}(1 - t_{\mathfrak{q}}) = 0$, so $A_{\mathfrak{q}} = 0$ for every $\mathfrak{q}$, and then $B = 0$ term by term. $\square$

Corollary 3.2 (The proposed generator is not invariant). *The choice $a_{\mathfrak{p}} = w_{\mathfrak{p}}$, $b_{\mathfrak{p}} = w_{\mathfrak{p}} N\mathfrak{p}^{-\beta}$ gives $b_{\mathfrak{p}} = a_{\mathfrak{p}} N\mathfrak{p}^{-\beta}$, violating (3.2): the sign of the exponent is reversed. Explicitly, with $t = N\mathfrak{p}^{-\beta}$ and $w_{\mathfrak{p}} = 1$: for every $\mathfrak{q} \neq \mathfrak{p}$ the proposed rates satisfy $A_{\mathfrak{q}} t_{\mathfrak{q}} + B_{\mathfrak{q}} = (t_{\mathfrak{q}}^{-1} - t_{\mathfrak{q}}) t_{\mathfrak{q}} + (t_{\mathfrak{q}}^2 - 1) = 0$, so the $\mathfrak{q}$ -terms of (3.1) cancel at $f = e_{\mathfrak{p}}$ and the value is carried by the $\mathfrak{p}$ -term alone:*

$$\varphi(\mathcal{L}(e_{\mathfrak{p}})) = t(t^{-1} - t) + t(t^2 - 1) = (1 - t^2)(1 - t) > 0$$

for $t \in (0, 1)$; numerically 0.5926 at $(N\mathfrak{p}, \beta) = (3, 1)$, 0.8779 at $(3, 2)$, 0.9033 at $(5, 1.5)$, 0.8397 at $(7, 1)$. With $b_{\mathfrak{p}} = a_{\mathfrak{p}} N\mathfrak{p}^{+\beta}$ the same expression is exactly 0.

Remark 3.3 (What the corrected ratio means). Condition (3.2) reads $b_{\mathfrak{p}}/a_{\mathfrak{p}} = e^{\beta \log N\mathfrak{p}}$: the backward rate exceeds the forward rate by the Boltzmann factor of the energy $\log N\mathfrak{p}$ transferred by $\mu_{\mathfrak{p}}$. This is quantum detailed balance in the sense standard for quantum Markov semigroups [1], and it is the only choice compatible with φ_{β} . The proposal's $N\mathfrak{p}^{-\beta}$ is the anti-detailed-balance choice, which drives the system away from equilibrium.

Note the price: (3.2) requires $\sum_{\mathfrak{p}} a_{\mathfrak{p}} N\mathfrak{p}^{\beta} < \infty$ for the second sum to converge in norm, so the weights must decay faster than $N\mathfrak{p}^{-\beta}$. That is a genuine constraint but is satisfiable, for instance by $a_{\mathfrak{p}} = N\mathfrak{p}^{-\beta-1-\epsilon}$. We have verified invariance, $\varphi \circ \mathcal{L} = 0$; full KMS-symmetry of the semigroup with respect to φ_{β} in the sense of Cipriani [2] is a stronger statement, which we have not checked.

4. NO CRITICAL SLOWING DOWN

Theorem 4.1 (Tensorisation to an infimum). *Let $\mathcal{L} = \sum_{\mathfrak{p}} a_{\mathfrak{p}} \mathcal{L}_{\mathfrak{p}}$ act on the product system $(V, \pi_{\beta}) = \bigotimes_{\mathfrak{p}} (\mathbb{N}, \mu_{t_{\mathfrak{p}}})$ with $t_{\mathfrak{p}} = N_{\mathfrak{p}}^{-\beta}$, each $\mathcal{L}_{\mathfrak{p}}$ the birth-death generator on its own factor with the corrected rates $(a_{\mathfrak{p}}, a_{\mathfrak{p}} t_{\mathfrak{p}}^{-1})$ of (3.2). Then:*

- (1) *the spectral gap satisfies $\text{gap}(\mathcal{L}) = \inf_{\mathfrak{p}} a_{\mathfrak{p}} \text{gap}(\mathcal{L}_{\mathfrak{p}}/a_{\mathfrak{p}})$, and by the Poincaré inequality for the geometric law (Paper IV, its Lemma 6.1, with sharp constant $C(t) = 4t/(1-t)^2$)*

$$\text{gap}(\mathcal{L}_{\mathfrak{p}}/a_{\mathfrak{p}}) \geq \frac{(1-t_{\mathfrak{p}})^2}{4t_{\mathfrak{p}}} \geq \frac{(1-t_{\mathfrak{p}})^2}{4};$$

- (2) *the modified log-Sobolev constant satisfies $\alpha(\beta) = \inf_{\mathfrak{p}} a_{\mathfrak{p}} \alpha_0(t_{\mathfrak{p}})$, where $\alpha_0(t)$ is the single-site constant of the normalized chain: entropy is subadditive over the product measure and the Dirichlet form is additive, giving $\geq$, and single-site test functions give $\leq$.*

Both constants are therefore infima of single-site quantities, each a function of its own $t_{\mathfrak{p}}$ alone — not products over $\mathfrak{p}$. In particular: with rates bounded below, $\inf_{\mathfrak{p}} a_{\mathfrak{p}} > 0$, the gap stays $\geq \inf_{\mathfrak{p}} a_{\mathfrak{p}} (1-3^{-\beta})^2/4 > 0$ uniformly for β in a neighbourhood of β_c ; with the norm-summable rates of Remark 3.3 the infima vanish for every β , for the weight reason alone and β -independently. In neither reading does any of these constants acquire the product shape $\Pi_{\infty}(\beta)$, whose vanishing at β_c is driven by the divergence of $\sum_{\ell} \ell^{-\beta}$, a tail quantity invisible to an infimum.

Proof. For commuting single-site generators on a product measure, the Dirichlet form and the variance resp. entropy split: $\text{Var} = \sum$ over sites of averaged single-site variances bounds the gap from below by the infimum, and single-site eigenfunctions attain it; the same subadditivity of entropy (a product measure satisfies $\text{Ent}(f) \leq \sum_{\mathfrak{p}} \mathbb{E} \text{Ent}_{\mathfrak{p}}(f)$) with the additive Dirichlet form gives $\alpha \geq \inf a_{\mathfrak{p}} \alpha_0(t_{\mathfrak{p}})$, and functions of a single coordinate give $\leq$. The single-site gap bound is Lemma 6.1 of Paper IV, whose Dirichlet form $\mathbb{E}_{\mu_t} |\Delta h|^2$ is that of the normalized chain with rates $(1, t^{-1})$. $\square$

Corollary 4.2 (The proposed asymptotic is of the wrong shape). $\Pi_{\infty}(\beta) = \prod_{\ell \in R} \frac{1-\ell^{-\beta}}{1+\ell^{-\beta}}$ is a product over all ℓ and tends to 0 as $\beta \downarrow \beta_c$, precisely because the series $\sum_{\ell} \ell^{-\beta}$ diverges there. By Theorem 4.1 the gap and MLSI constants are infima of single-site quantities, which do not degenerate this way: the single-site defect $1 - \ell^{-\beta}$ never falls below $1 - 3^{-\beta} \geq 0.66$, so with unit rates the gap infimum stays $\geq (1 - 3^{-\beta})^2/4 \geq 0.11$. Numerically, over the primes $3 \leq \ell \leq 10^4$,

β	$\inf_{\ell} (1 - \ell^{-\beta})$	Π_{∞}
1.5	0.8076	$3.70 \cdot 10^{-1}$
1.2	0.7324	$1.27 \cdot 10^{-1}$
1.05	0.6845	$3.58 \cdot 10^{-2}$
1.001	0.6670	$1.86 \cdot 10^{-2}$

The product collapses; the single-site defect does not fall below $1 - 3^{-\beta} \geq 0.66$. Hence $\alpha(\beta) \not\asymp \Pi_{\infty}(\beta)$, and there is no critical slowing down of this kind.

Corollary 4.3 (Bakry–Émery). *The same argument applies to the curvature-dimension condition: for product systems with additive generators the tensorised curvature bound is again an infimum of single-site quantities, each a function of its own $t_{\mathfrak{p}}$ alone. Whatever the single-site value, the bound therefore has no product shape, and no degeneration tied to β_c beyond what the weights impose β -independently: there is no Ricci-flat singularity at the transition.*

Remark 4.4 (Why the transition is invisible). The arithmetic phase transition is the Kakutani dichotomy: an infinite product of single-site affinities degenerates although every factor stays bounded away from 0 (Paper IV, its Lemma 5.1). It is therefore a *tail* event, detectable only by

quantities sensitive to infinitely many coordinates at once. Modified log-Sobolev constants and Bakry–Émery curvature are local: they tensorise, so they see only the worst single site. The same mechanism produced Theorem 6.2 of Paper IV, where the metric transition was continuous rather than first-order because the order parameter is a tail observable outside the Lipschitz domain, and Theorem 2.3 of Paper III, where Ξ_β was identified as an invariant of a measure class. Three different local invariants, one tail phenomenon, three failures to detect it.

5. ASSESSMENT

Proposal	Status	Reference
$[D_{\text{full}}, x]_\vartheta$ bounded	false: unbounded	Prop. 1.1
untwisted $[D_{\text{vert}}, \mu_a]$ bounded	true	Prop. 1.2
the twist is needed	no; it creates the problem	Rem. 1.3
$\ker L = \mathbb{C}1$	true iff S norm-separated	Prop. 2.1
$(\mathcal{A}, \mathcal{H}, D_{\text{full}})$ a spectral triple	not established: reduces to D_{base}	Rem. 2.2
$\mathcal{A}_{\mathbb{Q}, S}$ a compact quantum metric space	not established	Rem. 2.3
$\varphi \circ \mathcal{L}_\beta = 0$ as proposed	false: sign error	Cor. 3.2
$b_p = a_p \text{Np}^{+\beta}$	the correction	Prop. 3.1
KMS-symmetry (Cipriani sense)	not checked	Rem. 3.3
$\alpha(\beta) \asymp \Pi_\infty(\beta)$	false: infimum, not product	Thm. 4.1, Cor. 4.2
critical slowing down	does not occur	Cor. 4.2
$\kappa(\beta) \rightarrow 0$ at β_c	false	Cor. 4.3

Two of the three failures are repairable and the repairs are one-line: drop the twist, and flip the sign in the detailed-balance ratio. The third is not a repair but a structural statement, and it is the one worth keeping. Modified log-Sobolev and Bakry–Émery tensorise, hence are determined by the worst single coordinate, hence cannot detect a Kakutani dichotomy, which is by construction a statement about infinitely many coordinates whose individual contributions are all harmless; and the Connes distance of Paper IV is blind to the same transition in the complementary sense of its Theorem 4.2, the tail order parameter and all its truncations lying outside every Lipschitz domain. A quantum optimal transport theory that detects the arithmetic transition would have to be built from a genuinely non-tensorising quantity, and we do not know one.

Three questions. Is $(\mathcal{A}_{\mathbb{Q}, S}, L)$ with the untwisted D_{full} a compact quantum metric space, i.e. is the Lipschitz ball totally bounded? Does the corrected generator of Proposition 3.1 satisfy KMS-symmetry in the sense of Cipriani [2], and if so what is the associated non-tracial carré du champ? And is there any transport-theoretic quantity that fails to tensorise — the natural candidates being those built from the tail σ -algebra, where by [6, Prop. 7.2] the order parameter lives, rather than from a sum of single-site Dirichlet forms.

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